

Experiment 7

Program:-

```
#include <stdio.h>
#include <stdlib.h>
// Function to calculate FCFS
void fcfs(int arr[], int n, int head) {
    int seek = 0;
    for(int i = 0; i < n; i++) {
        seek += abs(arr[i] - head);
        head = arr[i];
    }
    printf("FCFS Total Seek Time: %d\n", seek);
}
```

```
// Function to calculate SCAN
void scan(int arr[], int n, int head, int size) {
    int seek = 0, i, j;
    // Sort array
    for(i = 0; i < n-1; i++) {
        for(j = 0; j < n-i-1; j++) {
            if(arr[j] > arr[j+1]) {
                int temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
}
```

```
// Move right
for(i = 0; i < n; i++) {
    if(arr[i] >= head) {
        seek += abs(arr[i] - head);
        head = arr[i];
    }
}
```

```
// Move to end
seek += abs(size - 1 - head);
head = size - 1;
// Move left
for(i = n-1; i >= 0; i--) {
    if(arr[i] < head) {
        seek += abs(arr[i] - head);
```

```

head = arr[i];
}
}

printf("SCAN Total Seek Time: %d\n", seek);
}
int main() {
int n, head, size;
printf("Enter number of requests: ");
scanf("%d", &n);

int arr[n];
printf("Enter requests:\n");
for(int i = 0; i < n; i++) {
scanf("%d", &arr[i]);
}
printf("Enter initial head position: ");
scanf("%d", &head);
printf("Enter disk size: ");
scanf("%d", &size);
fcfs(arr, n, head);
scan(arr, n, head, size);
return 0;
}

```

Output:-

Projects

My First Project

Help

\$0.00

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exp7.c

2026-03-24-file-1.term

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K

~\$ gcc exp7.c -o exp7

~\$./exp7

Enter number of requests: 5

Enter requests:

98 183 41 122 14

Enter initial head position: 50

Enter disk size: 200

FCFS Total Seek Time: 464

SCAN Total Seek Time: 334

~\$